



CHIRUNDU TOWN COUNCIL

PUBLIC HEALTH SECTION

**INTEGRATED SOLID WASTE MANAGEMENT PLAN
2024-2026**

1.0 INTRODUCTION

Background

Chirundu District shares a border with Zimbabwe, it has one (1) constituency with a total of 12 wards and an estimated population of 78, 780 with total number of households at 10, 528. Its population growth rate is at 4.3%. The district records temperature ranging from average of 19° C to 34° C.

Rationale

The ever increasing generation of waste in the district each day is one the most pressing issues today. Waste puts at risk public health, pollutes the environment and its disposal becomes a financial burden to the public. The management of various types of waste in District has been a very difficult and challenging issue. The challenge has manifested itself in the perpetual disease out breaks such as dysentery and pollution of water sources, air, soil, or land contamination, proliferation of pests and vermin, and the loss of aesthetic beauty.

Chirundu Town Council is responsible for providing waste management services and the management of municipal waste disposal facilities, and ensures, in liaison with other appropriate authorities.

The district is faced with a critical waste management problem, which is threatening the aspects of Health, Environment and socio-economic development. Major concerns include; Health hazards due to indiscriminate disposal of waste, littering which tempers with esthetic beauty, illegal dumping acting as rodent and bad odor, vermin breeding sites, Potential for contamination of soils and underground / surface water.

Goal

To promote an integrated solid waste management plan for the district in order to preserve human life and the environment.

Objectives

1. To promote reduction of waste generation through product minimization and reuse waste.
2. To promote reduction of waste disposed through product recovery/ recycling maximize the economic value of waste.
3. To provide proper waste transfer and disposal.
4. Develop and adopt environmentally sound treatment and disposal methods/practices.

2.0. POLICIES AND REGULATIONS

2.1. International conventions related to waste management

The Basel convention was established to address the global problem of uncontrolled movement and dumping of hazardous waste, including incidents of illegal dumping in developing countries by companies from developed countries. This was of great concern as indiscriminately disposed, accidental spillage or improper management of hazardous waste can pose severe health problems even death and can poison water and land for decades.

The Bamako convention is the convention on the ban of the import into Africa and control of Transboundary movements and Management of Hazardous wastes within Africa, which was adopted by the members of the African Union in 1991 and come into force in 1998.

2.2. National regulation framework of the plan

The Environmental Management Act No.12 of 2011 is the principal law on protecting the environment. It provides for the requirements for handling waste such as the licensing or permitting process for collection, transportation, treatment and disposal of waste. The Solid Waste Regulation and Management Act No. 20 of 2018 gives a provision for the sustainable regulation and management of solid waste, the incorporation of solid waste management companies and define their statutory functions. Additional supporting pieces of legislations with regards to waste management include the Local Government Act of 2019, Public Health Act Cap 295 of the Laws of Zambia.

3.0 THE STATE OF SOLID WASTE IN CHIRUNDU DISTRICT

The solid waste stream in the district comprises of the following; Domestic waste, Agriculture, E-waste, and industrial waste. Waste generated from all the sectors of the district are currently not well managed. The table below illustrates the solid waste stream distribution in approximation for Chirundu district.

	Category	Type	Source Details	Distribution
1	Municipal waste	Glass bottle	Beverage, food containers	High
		Organic	Vegetable,	Medium
		Plastics	Water, paper bags, bottles	High
2	Agriculture waste	Pesticide containers, stalks, stems, and leaves		low
3	Industrial waste	Masonry and concrete, scrap metal, oil, solvents, chemicals, scrap lumber and vegetable matter from restaurants.	Construction and Catering services	low
4	E-waste	Computers, Phones and household appliances		Trivial

KEY

High	Above 60%
Medium	Between 35-59 %
Low	Between 5-33%
Trivial	Below 4%

4.0 THE DISTRICT INTEGRATED SOLID WASTE MANAGEMENT PLAN

4.1. The need for the plan

The integrated solid waste management plan provides strategies in approaching sustainable management of solid waste in which the district will require to preserve the environment and human life. The growing concern of solid waste volume generated in Chirundu district put public health at risk, pollutes the environment and its disposal often cost the Local Authority a fortune it cannot afford. The formulation of a cost-effective management plan to deal with the current problem of poor waste management is a necessity.

4.2. Vision of the plan

The district solid waste management plan implementation will have to meet the following mandates;

- Establishment an integrated Solid waste management plan.
- Waste treatment and disposal facilities in the district.
- Promote of cost effective solid waste management.
- Promote Public awareness, education and communication.
- Legal and Regulatory framework to deal with producer responsibility among industry

The implementation of the plan will enhance the provision of solid waste management services, through infrastructure development, increased investment in equipment and capacity building. It is envisaged that various instruments, including bans, restrictions and taxations will be employed to deal with specific waste management system.

5.0 THE INTEGRATED SOLID WASTE MANAGEMENT PLAN

There are several positive action the district can take to improve waste management beyond waste collection and disposal. The plan is developed with the following components;

- Reduction of the amount of waste generated by reusing products.
- Reduce the amount of waste to be disposed by recycling or recovery.
- Pre-treatment/treatment and disposal of waste.

5.1 Development of a waste management systems

Ideally, a waste management system should consist of environmentally acceptable waste management practices that are aimed at minimizing solid waste generation and disposal. The system must provide for the protection of human health and the environment.

5.2 Waste management guiding Principles

The Principles provide guidance for both current solid waste generators and prospective solid waste management practitioners. The following principles are cardinal in the implementation of the plan.

Duty of care principle. The generator of waste should be responsible for all the fate of the waste generated in all circumstances. The generator should ensure that waste is handled, stored, collected, transported and disposed of according to legislation.

Cost sharing principle. Those who generate waste must bear the cost of its management and disposal. For too long, households have enjoyed local government subsidy for garbage collection and disposal. This situation has not helped develop among citizens a sense of responsibility for habits. It seems only fair that those who produce the wastes should shoulder part of it, if not the full cost of waste management.

Principle of Co-operation emphasizes on co-operation among all social groups is vital to solving environmental problems

Polluter pay principle causes the polluter or contaminator to be liable for any costs involved in cleaning it up the nuisance.

Cradle-to-grave principle - All stages of the production process are examined with regard to pollution.

Precautionary principle assumes that an unidentified contamination of waste is both highly hazardous and toxic until proven otherwise.

Waste is a resource and should be treat as an economic determinant.

Critical-load principle- limit of pollution, which the natural environment can withstand without becoming permanently damaged.

Integrated life-cycle principle- substances and products should be designed and managed in such a way that environmental impacts are minimized during generation, use, recovery and disposal

BAT principle -Best Available Technology- poor technical solutions are no excuse for pollution if better technology is available.

Waste prevention is better than waste regulation and control.

Preventing waste from being produced at all is much better than trying to manage it after a lot of it has been generated. With waste avoidance, reuse, recycling and materials recovery, we end up with less waste to collect, treat and dispose of.

There is no single management and technological approach to solid waste. An integrated solid waste management system will best achieve solid waste management goals.

Waste is generated in different types which can be managed and disposed of separately. The waste stream is made up of many distinct components, such as biodegradable, non-biodegradable, reusable, recyclable, toxic and hazardous, and each requires specific handling and disposal methods.

6.0 ISWM PLAN ACTIONS AND MEASURES

6.1.0. Reduction of solid waste generated and disposed

Source reduction

Source Reduction as an element of ISWM, represents an activity which actually precedes waste management, at the point of waste generation, by examining how products are manufactured, purchased and used.

It shall be the policy of the Local Authority of Chirundu town council to develop and manage a district source reduction program. This will be a comprehensive program to address all areas of source reduction including decreased consumption, reduced material volume, reduced toxicity or products and/or operating procedures, and reuse. The program is intended to make waste prevention a priority for Chirundu district with these goals;

- Develop programs to encourage residents to consider opportunities such as changing buying habits to reduce the purchase of single-use and disposable products wherever appropriate.
- Develop technical guidelines for management of different hazardous waste streams for commercial facilities
- Expand existing educational programs to encourage residents to implement source separation or composting as an alternative to the disposal of yard waste.
- Develop programs to assist businesses with source reduction efforts.
- Improve on waste pre-treatment status of currently running incinerators – upgrade the existing incinerators in particular at medical institutions.
- Open up existing central incinerators to allow smaller clinics and health centers to use facilities for incineration.

Recycling

Recycling has always been an option in waste management systems, but was not commonly implemented on a large scale as a waste management tool. Recycling consists of separating materials from the waste stream, either at the source, or by recovering materials from mixed waste prior to disposal.

The Local Authority to implement programs to recover recyclable materials from the solid waste stream within Chirundu district. Recycling programs will include both materials mandated for recycling by law or regulation, and any other materials which may be effectively recovered and marketed as commodities. The following shall be the goals;

- Encourage the procurement by local businesses, government and residents of all manner of goods made all or in part from recycled materials.
- Provide for a reduction in the waste stream requiring ultimate disposal by recovering recyclable materials to the maximum extent practical.
- To help in the development and expansion of secondary materials markets at the local, regional, state and national levels.
- Provide for a recycling program convenient for all residents and businesses in Chirundu district to encourage source separation of materials from the waste stream.
- Provide for sufficient processing capacity for source-separated recyclable materials generated.

Landfill

The Local Authority shall utilize landfill/dumpsite disposal for the wastes and residues remaining after applying waste reduction, recycling, organics reclamation and combustion technologies to the waste stream.

The Local Authority shall provide sufficient dumpsite disposal capacity to assure the continued safe and environmentally sound management of solid waste in the event that waste reduction, recycling, organics reclamation or combustion technologies do not achieve their full potential for reduction of the waste stream. The following shall be the goals;

- Utilize landfill designs and construction practices which assure compliance with regulatory requirements by ZEMA and provide long term protection of the environment.
- Implement cost effective programs to recover usable and marketable materials from the waste received at the landfill, prior to disposal.
- Review and refine operating practices on an ongoing basis to assure the maximal use of available landfill capacity.
- Evaluate landfill consumption and capacity on an annual basis to determine the remaining life of disposal facilities and to ensure that policy decisions regarding the expansion or replacement of landfill capacity are addressed in a timely manner.

Collection Services

Policy Statement

The Local Authority of Chirundu District shall provide for an adequate level of solid waste and recycling collection for the residents and businesses. It shall assure that collection services are dependable and cost effective. The following are the goals;

- Provide a level of service which meets the collection needs of the vast majority while encouraging recycling and the separation of vegetative waste.
- Promote product recovery.
- Utilize a competitive process to select competent collectors to provide collection services and assure exclusivity of franchise collection contracts as required.
- Assure compliance with regulatory requirements for sanitary collection and transport of wastes.

Public participation

Chirundu town council shall provide public information and education programs for all components of the integrated solid waste management system and to inform the public and industry organizations of the systems objectives.

- Develop comprehensive educational programs for all the elements of the integrated solid waste management system.
- Develop new and enhance existing youth/school education programs.
- Provide forums for the public such as media and public gatherings to present concerns and recommendations and involve community and business interests in early planning efforts when programs being developed affect them financially.
- Prepare audio/visual, print productions and personal presentations that enhance the achievement of the educational objects.

7.0 EXISTING CONDITIONS

7.1 Solid Waste System Capacities and Services.

The solid waste system developed and operated by the Local Authority consists of transfer and disposal facilities which the Authority owns or monitors. In addition to the facilities of the Authority, certain transfer and disposal facilities owned and operated by units of sub-contracted franchise solid waste companies.

Solid waste is collected from the Business premises unit and House-Hold units. There are (3) three groups of solid waste collectors;

- Waste-Cart subunit,
- Tractor subunit,
- Tipper-Front End Loader and Area cleaning border.

The waste cart subunit shall operate daily within the market ferrying solid waste to the nearest bay from the small shops and market shelters; The Tractor subunit shall collect from paid Business premises within Central Business District; The Tipper and front end loader shall collect twice a week from bays whereas the area cleaning border subunit shall service the border area and sweep the T2 road, Hospital road and Chiawa road.

Sub-contract

The local Authority to sign an agreement between Franchise solid waste companies, these shall be mandated to collect House-hold solid waste on behalf of the Local Authority within specified area zones. A franchise will meet a minimum requirement the contract will need and is regulated by Chirundu town council.

8.0 ACTIONS AND MEASURES TO BE EMPLOYED TO DEAL WITH GENERATION PATTERNS

8.1 Storage

Because solid waste might not necessarily have to be disposed of as soon as it is generated, the storage of it becomes critical. The storage of waste at an appropriate site provides for the planning of the frequency of collection. Actions and measures to improve storage of waste will be to:

- Improve on appropriate management systems for transfer station which will enable prevention of nuisance generation.
- Mobilize financial resources for procurement of storage equipment such as public waste bins and skip bins.
- Encourage separate storage of waste of different nature and composition to enhance recovery of useful materials and prevent cross contamination.
- Store waste according to the set conditions as provided for in the regulations governing the management of both hazardous and non-hazardous waste.

8.2 Infrastructural development

To ascertain that the management of solid waste is enhanced, the development of waste management facilities such as disposal sites will be key. The Local Authority shall embark to do the following;

- Promote private public sector and community partnerships.

- Encourage investment in waste management infrastructural development.
- Develop an enabling environment for investment in waste management.

9.0 INSTITUTIONAL FRAMEWORK

The institutional framework will include Government institution and agencies business and industry, Non-Governmental Organization and the public. The Local Authority will have to continue executing the responsibility conferred on them by law of ensuring planning and provision of collection and disposal services for Solid Waste. Private industry will need to be encouraged to actively participate in the management of waste. There has to be built both financial and technical capacity for all key players in waste management.

10.0 MONITORING AND EVALUATION

The Local Authority shall engage in regular monitoring to institute check-up on the progress in the implementation of the plan. Reports shall complement evaluation.

10.1 Evaluation

The evaluation of the plan will focus on assessing the progress of implementing the required improvements, and how far the objectives are being achieved through Government and Public sector as well as community involvement.

11.0 IMPLEMENTATION OF THE PLAN

Chirundu Town Council will spearhead and coordinate the implementation of the collaboration with other relevant stakeholders such as institutions of learning and NGOs. The Local Authority will launch the plan and ensure that plan is implemented. The implementation will include development of specific work plans from the broader targets provided within this plan and also taking into consideration the time frames.

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Date 06 / March/ 2024

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